

SANTIAGO NEIRA LOPEZ

University of Massachusetts Amherst • Department of Mathematics, LGRT 1235M
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PROFESSIONAL SUMMARY

PhD student in Mathematics specializing in statistics, machine learning, and scientific computing. Experienced in developing probabilistic models, deep learning methods, and numerical algorithms for large-scale scientific applications. Interested in Machine Learning, Data Science, and Quantitative Research internships.

EDUCATION

PhD in Mathematics <i>University of Massachusetts Amherst</i> GPA: 3.7 / 4.0	September 2023 – Present
MSc in Mathematics <i>University of Massachusetts Amherst, MA, USA</i> GPA: 3.8 / 4.0	September 2023 – September 2025
BSc in Mathematics <i>Pontificia Universidad Javeriana, Bogotá, Colombia</i> GPA: 4.6 / 5.0	January 2019 – February 2023

TECHNICAL SKILLS

- **Programming:** Python, R, SQL, Git
- **Libraries:** NumPy, SciPy, Pandas, Scikit-Learn, PyTorch, GPy, Emukit
- **Machine Learning:** Linear Models, Gaussian Processes, Bayesian Optimization, Deep Learning, Operator Learning, Generative Models, Probabilistic Machine Learning
- **Statistics:** Statistical Inference, Bayesian Statistics, Probability Theory, Stochastic Processes
- **Applied Mathematics:** Numerical Methods, Optimization, PDEs

EXPERIENCE

Graduate Teaching Assistant <i>University of Massachusetts Amherst</i>	September 2023 – Present
• Led weekly discussion sections for 30–40 undergraduate students in Calculus I and II. • Designed problem-solving sessions and provided one-on-one support, improving students' understanding of core mathematical concepts. • Communicated complex mathematical ideas effectively to students with diverse backgrounds.	

PROJECTS

Heston Stochastic Volatility Model <i>Collaborative Project</i>	Python, NumPy, SciPy, Pandas
• Implemented European option pricing algorithms under the Heston stochastic volatility model in Python. • Developed a calibration pipeline combining global and local optimization methods. • Applied the calibration framework to real TSLA options market data. • Benchmarked pricing accuracy and computational efficiency across numerical methods. • Repository: Available on GitHub.	

CERTIFICATIONS & COURSEWORK

- **Certifications:** Summer 2026 Quant Finance Boot Camp — The Erdős Institute (2026)
- **Selected Courses:** Machine Learning, Bayesian Statistics, Statistical Inference, Probability, Stochastic Processes, Numerical Methods, Mathematics of Generative AI

HONORS & AWARDS

- **2023:** Orden al Mérito Académico Javeriano
- **2023:** Best GPA, Honors for Bachelor's Thesis